# Student Perfomance Factors - Dataset analysis

## 1 Data statistics, distribution and interpretation

### 1.1 Statistics

The dataset consists of 20 features of both numerical and categorical types. Present categorical features are ordinal and nominal. Target label is learning disabilities. In the following table, some statistical results were computed for all features.

| **Statistic** | **Hours Studied** | **Attendance** | **Parental Involvement** | **Access to Resources** |
|---|---|---|---|---|
| Count | 6377 | 6377 | 6377 | 6377 |
| Unique | - | - | 3 | 3 |
| Top | - | - | Medium | Medium |
| Freq | - | - | 3251 | 3203 |
| Mean | 19.98 | 80.02 | - | - |
| Std | 5.99 | 11.55 | - | - |
| Min | 1.00 | 60.00 | - | - |
| 25% | 16.00 | 70.00 | - | - |
| 50% | 20.00 | 80.00 | - | - |
| 75% | 24.00 | 90.00 | - | - |
| Max | 44.00 | 100.00 | - | - |

Table 1: Summary Statistics for Hours Studied, Attendance, Parental Involvement, and Access to Resources

| **Statistic** | **Extracurricular Activities** | **Sleep Hours** | **Previous Scores** | **Motivation Level** |
|---|---|---|---|---|
| Count | 6377 | 6377 | 6377 | 6377 |
| Unique | 2 | - | - | 3 |
| Top | Yes | - | - | Medium |
| Freq | 3806 | - | - | 3237 |
| Mean | - | 7.04 | 75.06 | - |
| Std | - | 1.47 | 14.40 | - |
| Min | - | 4.00 | 50.00 | - |
| 25% | - | 6.00 | 63.00 | - |
| 50% | - | 7.00 | 75.00 | - |
| 75% | - | 8.00 | 88.00 | - |
| Max | - | 10.00 | 100.00 | - |

Table 2: Summary Statistics for Extracurricular Activities, Sleep Hours, Previous Scores, and Motivation Level

| Statistic | Internet Access | Tutoring Sessions | Family Income | Teacher Quality |
|---|---|---|---|---|
| Count | 6377 | 6377 | 6377 | 6377 |
| Unique | 2 | - | 3 | 3 |
| Top | Yes | - | Low | Medium |
| Freq | 5893 | - | 2582 | 3826 |
| Mean | - | 1.49 | - | - |
| Std | - | 1.23 | - | - |
| Min | - | 0.00 | - | - |
| 25% | - | 1.00 | - | - |
| 50% | - | 1.00 | - | - |
| 75% | - | 2.00 | - | - |
| Max | - | 8.00 | - | - |

Table 3: Summary Statistics for Internet Access, Tutoring Sessions, Family Income, and Teacher Quality

| Statistic | School Type | Peer Influence | Physical Activity | Learning Disabilities |
|---|---|---|---|---|
| Count | 6377 | 6377 | 6377 | 6377 |
| Unique | 2 | 3 | - | 2 |
| Top | Public | Positive | - | No |
| Freq | 4433 | 2552 | - | 5709 |
| Mean | - | - | 2.97 | - |
| Std | - | - | 1.03 | - |
| Min | - | - | 0.00 | - |
| 25% | - | - | 2.00 | - |
| 50% | - | - | 3.00 | - |
| 75% | - | - | 4.00 | - |
| Max | - | - | 6.00 | - |

Table 4: Summary Statistics for School Type, Peer Influence, Physical Activity, and Learning Disabilities

| Statistic | Parental Education Level | Distance from Home | Gender | Exam Score |
|---|---|---|---|---|
| Count | 6377 | 6377 | 6377 | 6377 |
| Unique | 3 | 3 | 2 | - |
| Top | High School | Near | Male | - |
| Freq | 3158 | 3801 | 3688 | - |
| Mean | - | - | - | 67.25 |
| Std | - | - | - | 3.89 |
| Min | - | - | - | 55.00 |
| 25% | - | - | - | 65.00 |
| 50% | - | - | - | 67.00 |
| 75% | - | - | - | 69.00 |
| Max | - | - | - | 100.00 |

Table 5: Summary Statistics for Parental Education Level, Distance from Home, Gender, and Exam Score

## 1.2 Distribution

Shapiro-Wilk test is used to verify if the dataset is normally distributed. Categorical features were excluded. The following values were obtained:

| Variable | Shapiro-Wilk Statistic | Shapiro-Wilk p-value |
|---|---|---|
| Hours_Studied | 0.997451 | 6.829164e-09 |
| Attendance | 0.954746 | 1.475918e-40 |
| Sleep_Hours | 0.952878 | 3.520622e-41 |
| Previous_Scores | 0.955414 | 2.493989e-40 |
| Tutoring_Sessions | 0.889654 | 0.000000e+00 |
| Physical_Activity | 0.920932 | 0.000000e+00 |
| Exam_Score | 0.905537 | 0.000000e+00 |

Table 6: Shapiro-Wilk Statistics and p-values for numerical variables

For most features, the Shapiro statistic is close to 1, which could mean they are close to a normal distribution. But because the p-value for all features is below 0.05, they do not abide to a normal distribution. The Shapiro-Wilk test is sensitive for large datasets (bigger than 5000), as is in this case. So, small deviations from the normal distribution can lead to the test detecting a significant difference.

### 1.3 Interpretation

The values computed for numerical features are: count, mean, standard deviation, min, 25, 50, 75 percentile and max.

For hours studied, most entries are between 16 and 24, centered around 19.98. There are outliers due to the min value being 1 and max being 44, with an emphasis towards the max. The distribution is right skewed.

For attendance, most entries are between 70 and 90 centered around 80.

For sleep hours, entries are between 6 and 8. With standard deviation 1.47 and 7.04 mean, most entries are present in 6-8 range, with outliers between 4-6 and 8-10.

For previous scores, most entries are between 63 and 88 with min/max values close to these values, suggesting a normal distribution. They are centered around 75.

For tutoring sessions, most entries are between 1 and 2 and centered around 1.49. Max value is 8, suggesting a right skewed distribution.

For physical activity, most entries are between 2 and 4 and centered around 3.

For exam score entries fall between 65 and 69 centered around 67 with a slight deviation of 3.89. With the mean being almost at the center of the interval made by the 25% and 75% quartiles, most entries are consistently in this range.

The values computed for categorical features are: count, unique, top, frequency. Unique represents the number of unique values present in said column. Top represents the most occurring value. Freq represents the number of occurrences of top. For categorical features like learning disabilities and internet access, we observe high data imbalance. The most frequent value for learning disabilities is present in 89% of entries, and for internet access it is present in 92% of entries. This can lead the model to overfit the majority class, while underfitting the minority class.

## 2 Feature analysis

All categorical features shall be encoded using label encoding. Given the nature of this dataset, both normal and non-normal distributions will be explored.

A first step in identifying redundant features is to use a Pearson correlation matrix for linear dependencies identification. The heat map corresponding to the matrix is the following:

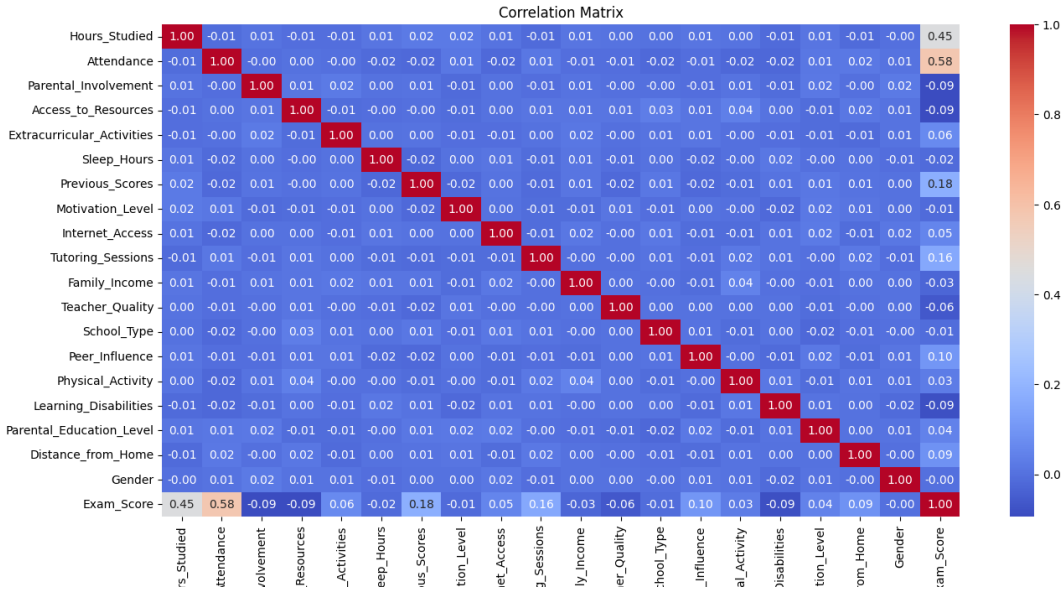


Figure 1: Pearson correlation matrix heat map

Diagonal coefficients are 1 because each feature is correlated with itself. Most features are nearly independent of one another with coefficients close to 0. Exam score is the exception, presenting a linear dependence with attendance. Other weak linear dependencies are present. Because of the lack of linear dependencies to our target label, we cannot select relevant features.

| **Feature** | **p-value** |
|---|---|
| Parental_Involvement | 0.5984 |
| Access_to_Resources | 0.8647 |
| Extracurricular_Activities | 0.3959 |
| Motivation_Level | 0.2176 |
| Internet_Access | 0.7341 |
| Family_Income | 0.5201 |
| Teacher_Quality | 0.9678 |
| School_Type | 0.8685 |
| Peer_Influence | 0.5089 |
| Parental_Education_Level | 0.4499 |
| Distance_from_Home | 0.8946 |
| Gender | 0.1007 |

Table 7: Independence Tests between Categorical Features and Learning Disabilities using Pearson's Chi Square

Pearson Chi-Squared test is used to determine if there is a significant association between two categorical features. A p-value larger than the threshold of 0.05 denotes a feature independent of the target label. This reinforces the conclusions from the correlation matrix computed earlier.

| Feature | Kendall's Tau |
|---|---|
| Parental Involvement | -0.011756 |
| Access to Resources | 0.005657 |
| Motivation Level | -0.018374 |
| Family Income | -0.003404 |
| Teacher Quality | -0.000459 |
| Peer Influence | -0.013682 |
| Parental Education Level | 0.005548 |
| Distance from Home | 0.002852 |

Table 8: Kendall's Tau values for ordinal features

Kendall Tau is a test of association which quantifies the relationship between 2 ordinal features. Test range is between -1 and 1 and denotes the direction of the relationship and its strength. A value close to 0 denotes a weak relationship. A positive Kendall Tau coefficient represent a direct proportionality, while a negative value indicates inverse proportionality.

We can use the F-Test to evaluate whether there is a statistically significant difference in the target variable across different values or categories of the feature. A high f-score denotes a strong association with the target feature.

| Feature | F-score | p-value |
|---|---|---|
| Hours Studied | 1.22 | 0.26849 |
| Attendance | 2.76 | 0.09677 |
| Parental Involvement | 1.01 | 0.31587 |
| Access to Resources | 0.24 | 0.62177 |
| Extracurricular Activities | 0.79 | 0.37320 |
| Sleep Hours | 1.27 | 0.25890 |
| Previous Scores | 0.19 | 0.66051 |
| Motivation Level | 2.03 | 0.15390 |
| Internet Access | 0.17 | 0.67684 |
| Tutoring Sessions | 0.30 | 0.58415 |
| Family Income | 0.15 | 0.69772 |
| Teacher Quality | 0.00 | 0.95167 |
| School Type | 0.04 | 0.83376 |
| Peer Influence | 1.29 | 0.25613 |
| Physical Activity | 0.85 | 0.35604 |
| Parental Education Level | 0.17 | 0.68257 |
| Distance from Home | 0.02 | 0.88067 |
| Gender | 2.83 | 0.09242 |
| Exam Score | 45.24 | 0.00000 |

Table 9: F-scores and p-values of various features

A high f-score and a low p-value denotes a feature with strong predictive power. High f-score represents a strong relationship with the target label. Low p-value indicates the result is statistically significant. The exam score attribute complies with both requirements of a relevant feature. While the rest have lower f-score, they can still contribute to the prediction.

| Feature | Kruskal-Wallis p-value |
|---|---|
| Hours_Studied | 2.418306e-01 |
| Attendance | 9.876540e-02 |
| Parental_Involvement | 3.234026e-01 |
| Access_to_Resources | 6.346074e-01 |
| Extracurricular_Activities | 3.731560e-01 |
| Sleep_Hours | 1.857725e-01 |
| Previous_Scores | 6.594140e-01 |
| Motivation_Level | 1.227492e-01 |
| Internet_Access | 6.768110e-01 |
| Tutoring_Sessions | 9.715505e-01 |
| Family_Income | 7.746392e-01 |
| Teacher_Quality | 9.697059e-01 |
| School_Type | 8.337421e-01 |
| Peer_Influence | 2.490737e-01 |
| Physical_Activity | 2.858846e-01 |
| Parental_Education_Level | 6.410671e-01 |
| Distance_from_Home | 8.136343e-01 |
| Gender | 9.241813e-02 |
| Exam_Score | 2.371953e-14 |

Table 10: Kruskal-Wallis p-values for various features

Kruskal-Wallis test is used for datasets with non-normal distribution. The test compares the distributions of a numerical variable across two or more independent groups. It can determine if there are statistically significant differences between groups with respect to a target variable.

Low p-value denotes statistical significance. Similarly to F-Test, exam score is the most relevant feature.

# 3 Data visualization

Below we can visualize the distribution of the numerical features.

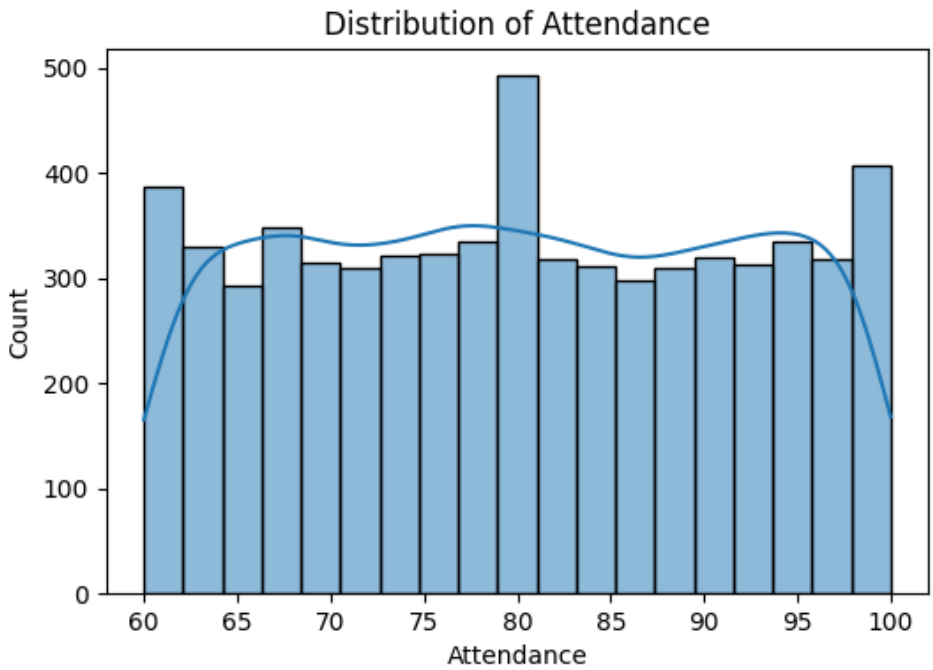


Figure 2: Attendance distribution

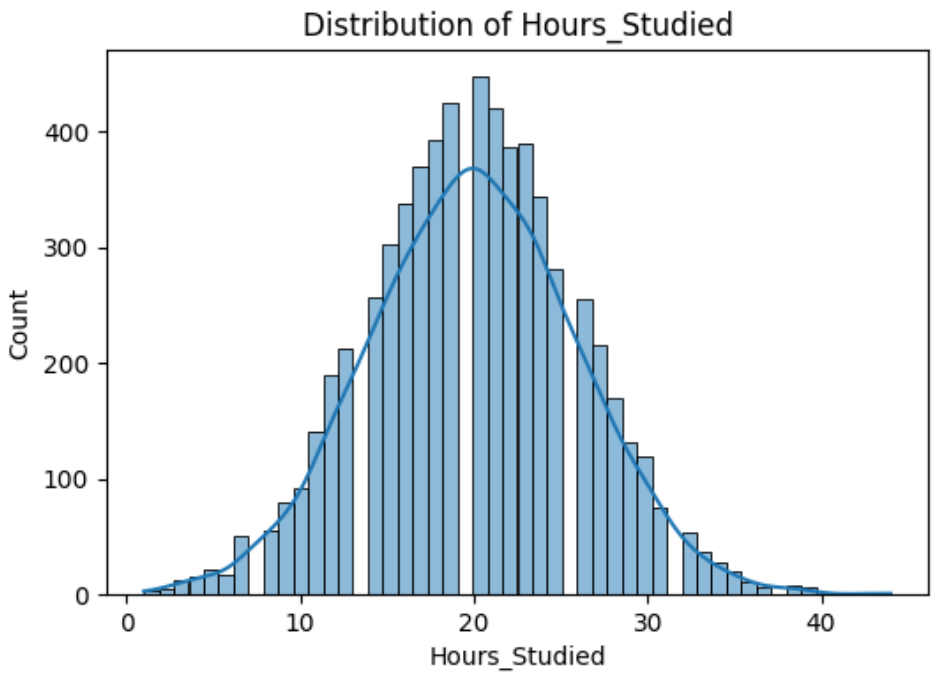


Figure 3: Hours studied distribution

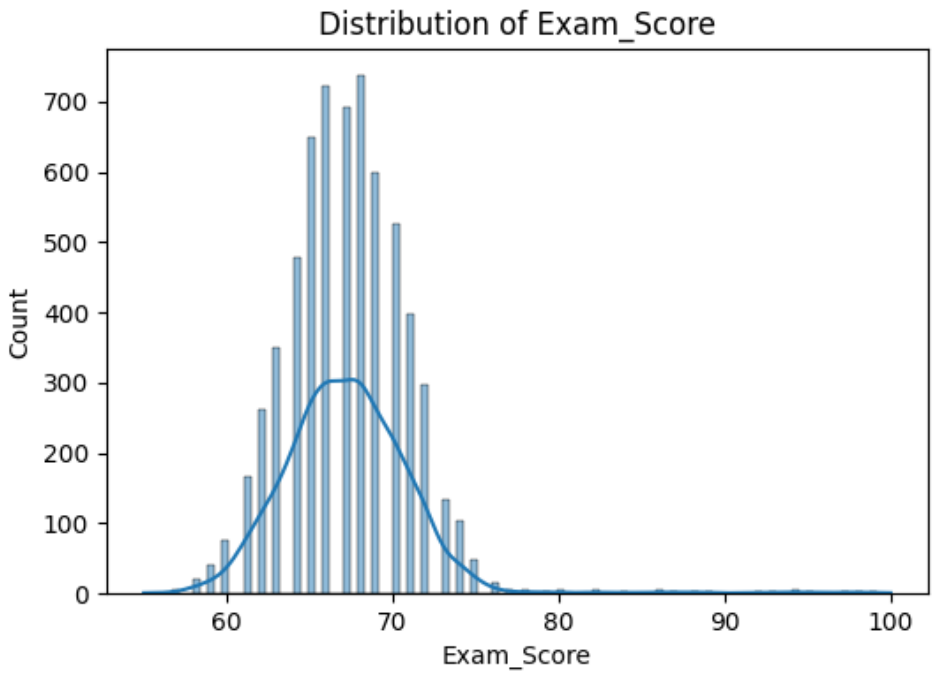


Figure 4: Exam score distribution

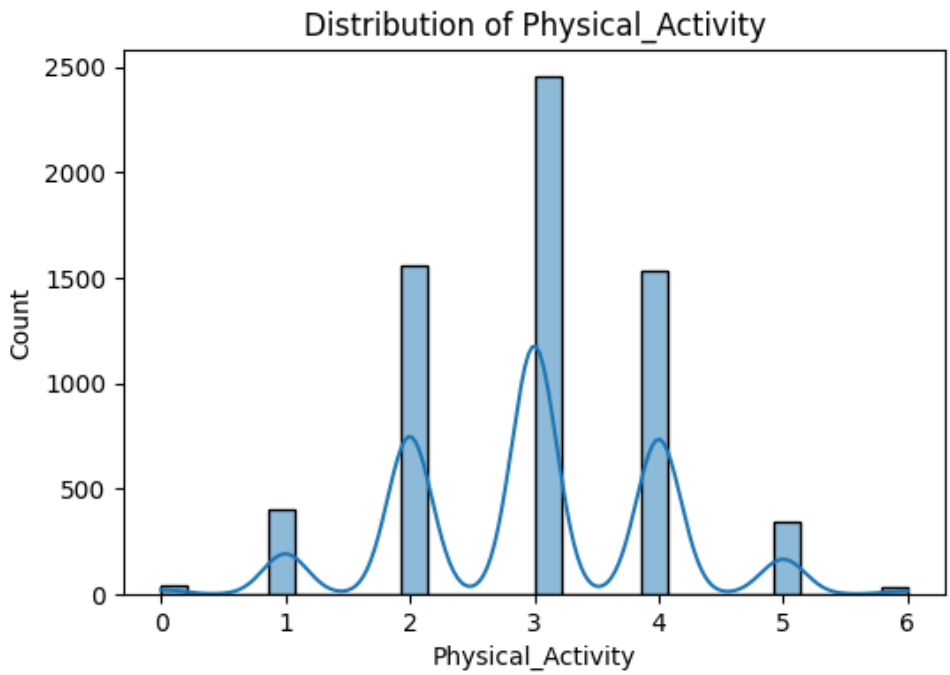


Figure 5: Physical activity distribution

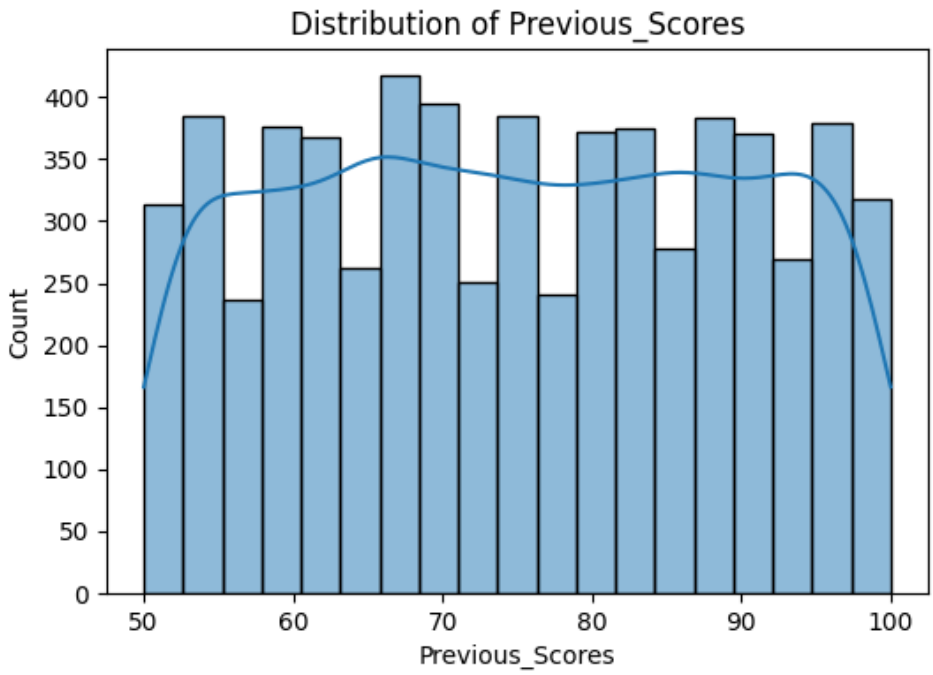


Figure 6: Previous scores distribution

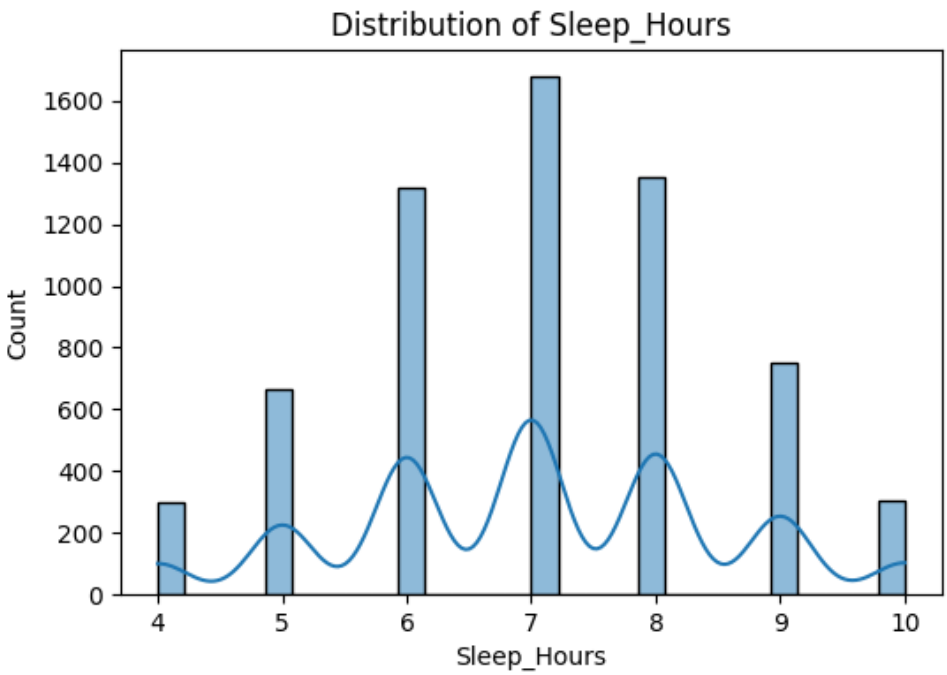


Figure 7: Sleep distribution

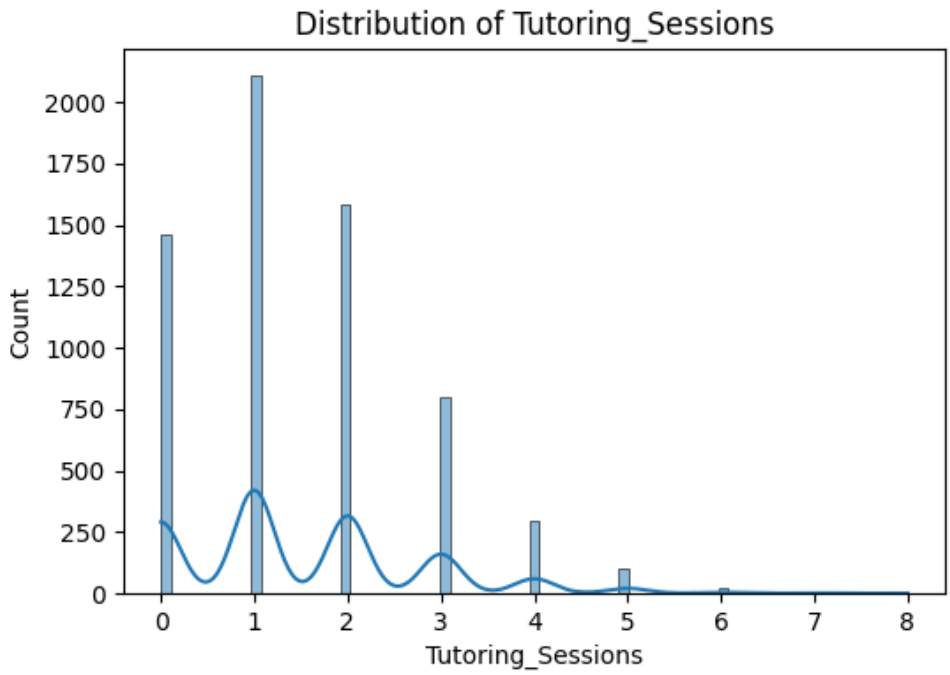


Figure 8: Tutoring distribution

For attendance, the distribution is near uniform, but with a concentration at 80.

For exam score, the distribution is right skewed, with most values clustering between 60 and 70.

Hours studied presents a close to normal distribution, having a slight skew to the right.

Physical activity has a normal distribution, with emphasis on 2, 3 and 4.

Previous scores is normally distributed.

Sleep hours has a normal distribution, with emphasis on 6, 7 and 8.

Tutoring sessions is right skewed, most entries being between 0 and 3, reinforcing the interpretation of said feature from the statistics table.

The next histograms represent the distribution of the categorical features.

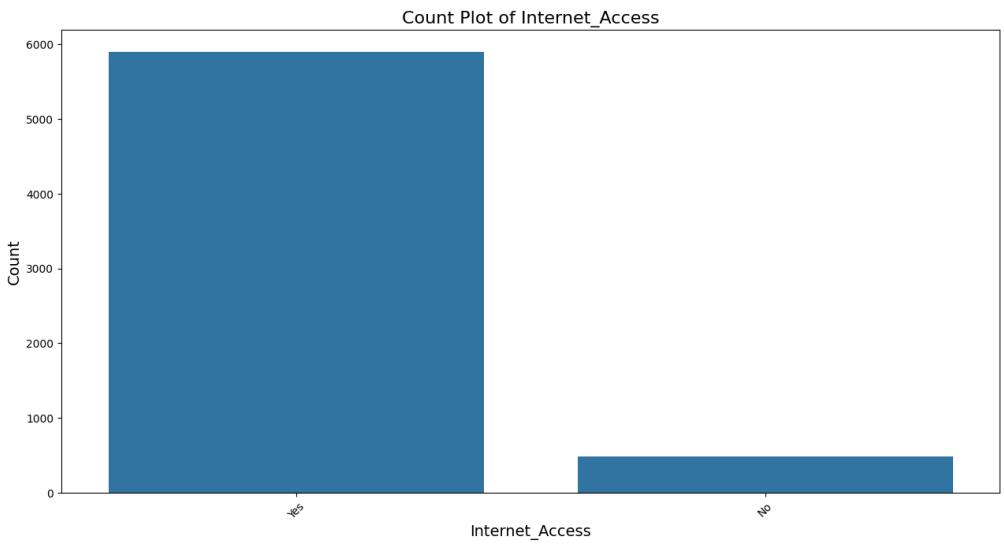


Figure 9: Internet access count

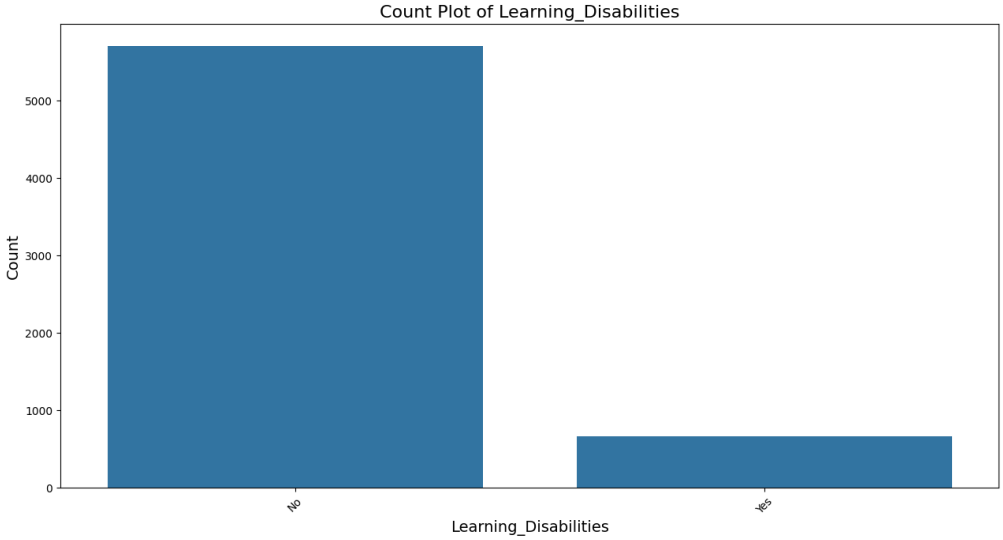


Figure 10: Learning disabilities count

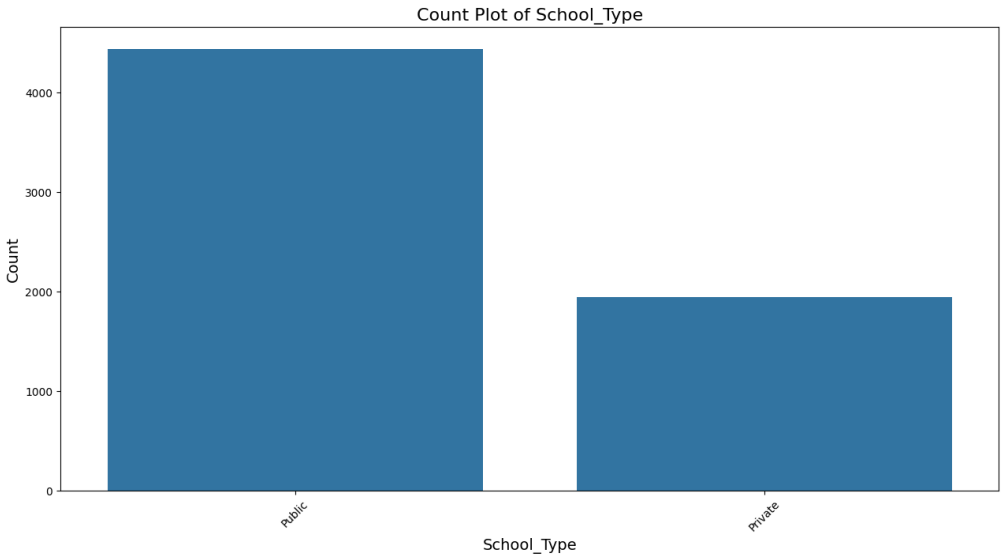


Figure 11: School type count

We can observe the imbalance of learning disabilities and internet access, as described in the Data Statistics chapter.

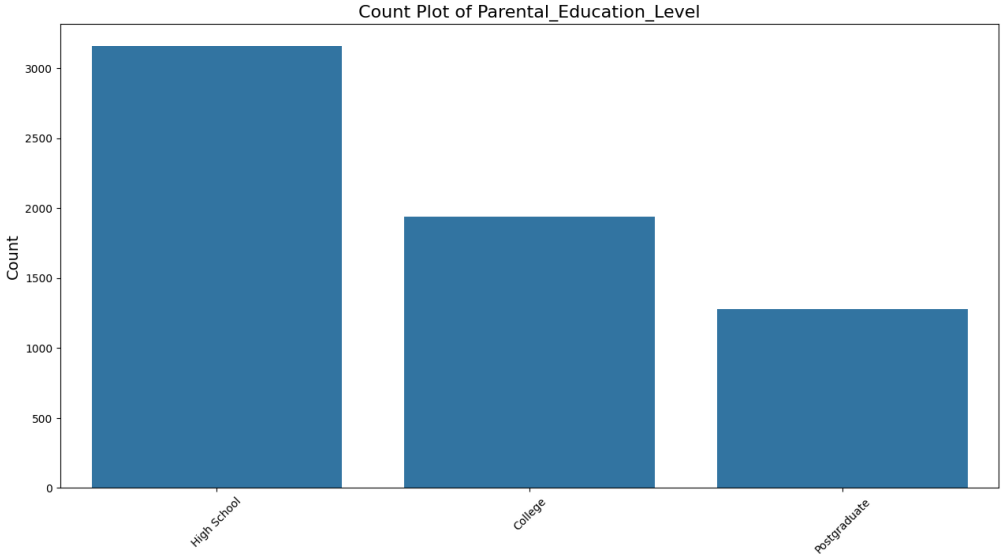


Figure 12: Parental education count

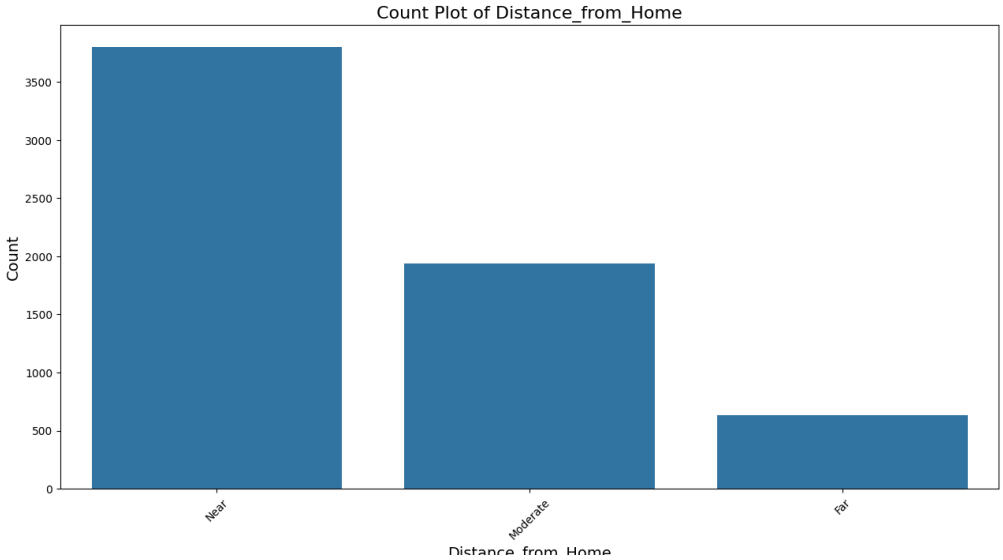


Figure 13: Distance from home count

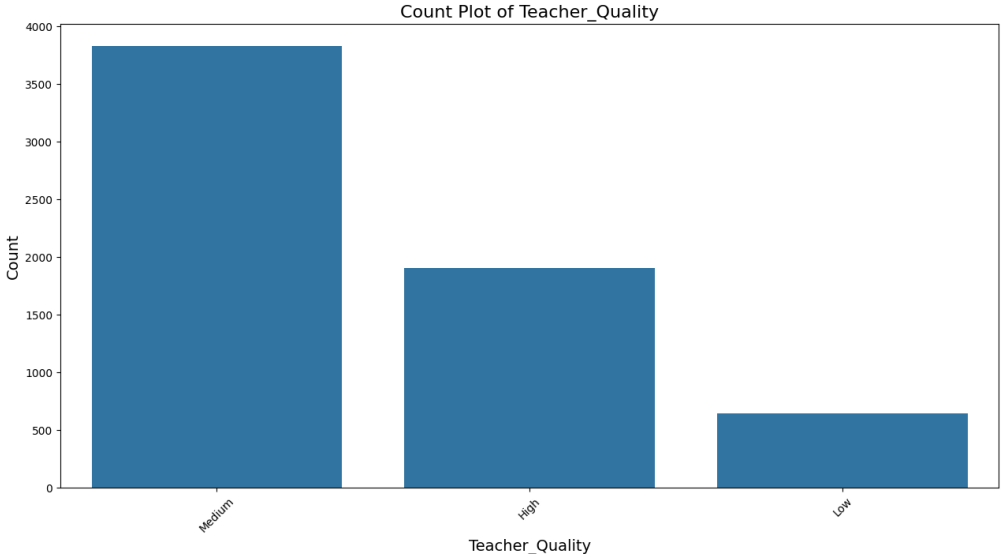


Figure 14: Teacher quality count

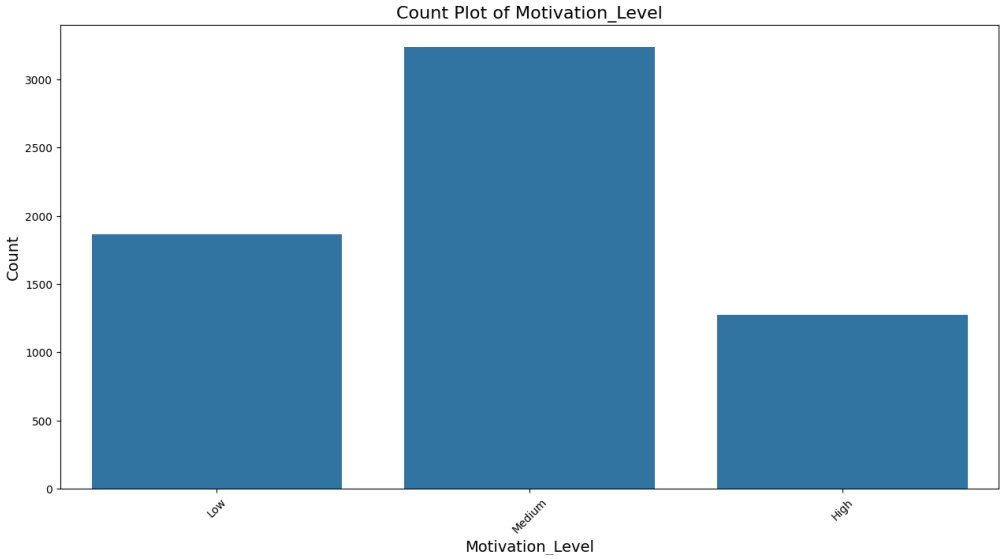


Figure 15: Motivational level count

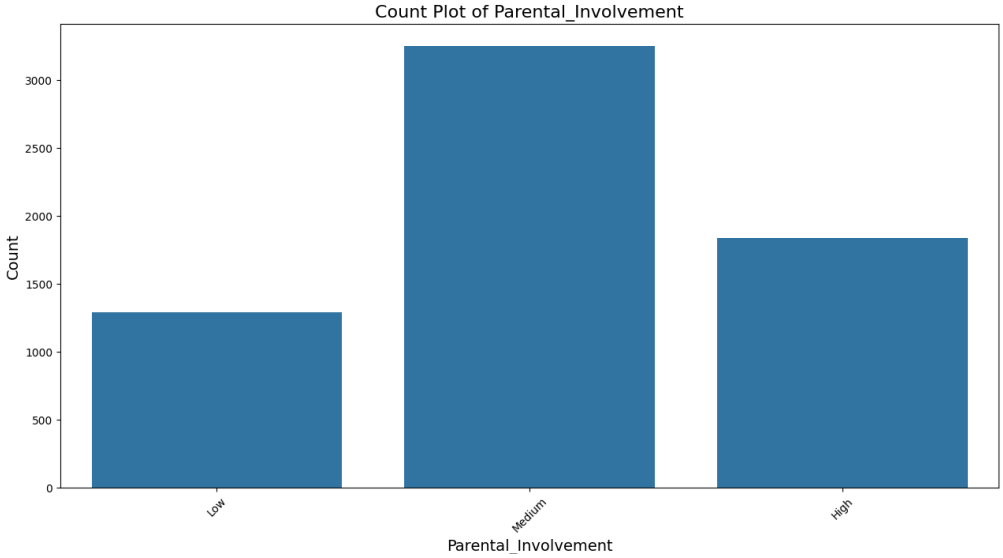


Figure 16: Parental involvement count

Parental education level is better balanced, with more entries in High School category. Distance from home shows a slight imbalance, having the majority of entries in near, while the far category is under-represented. Teacher quality, parental involvement and motivational level show a similar pattern as Distance from home.

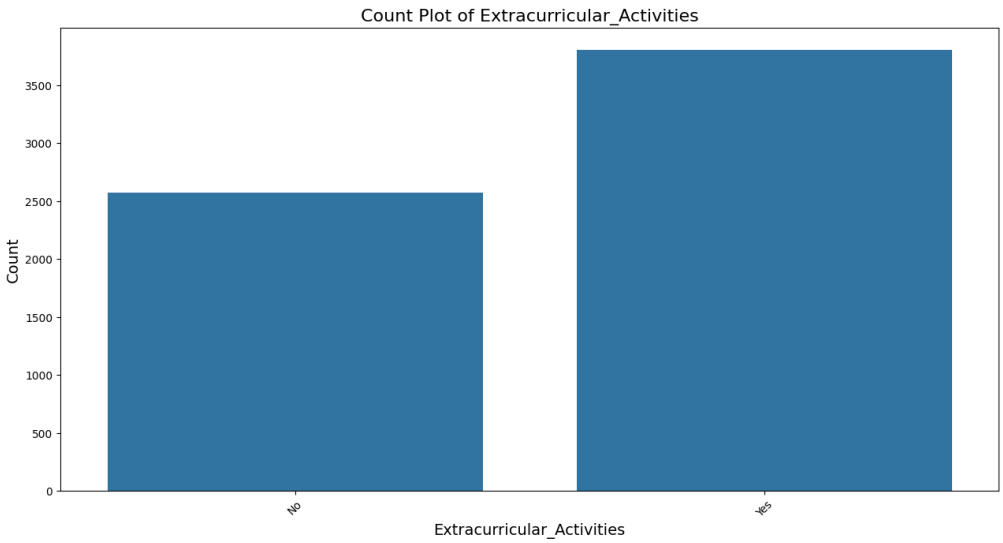


Figure 17: Extracurricular activity count

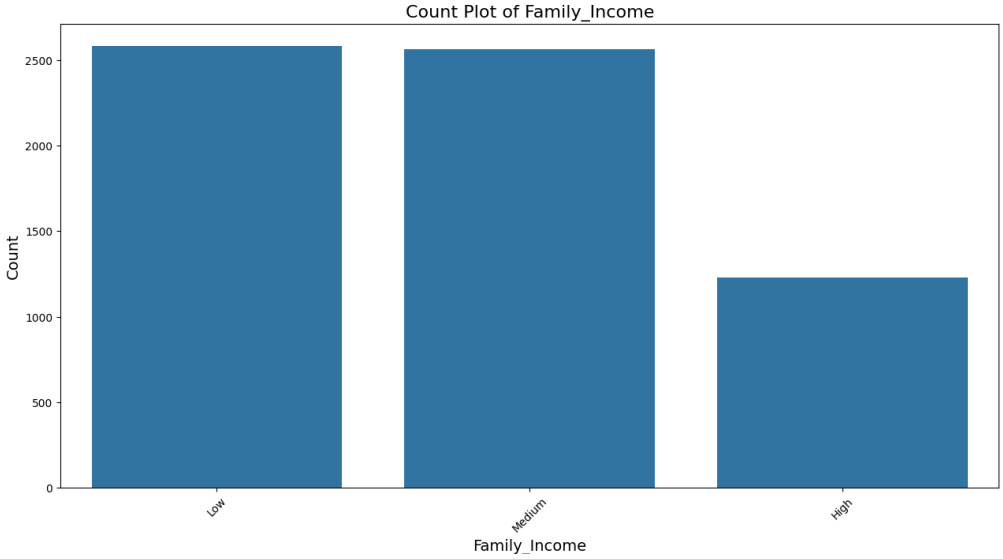


Figure 18: Family income count

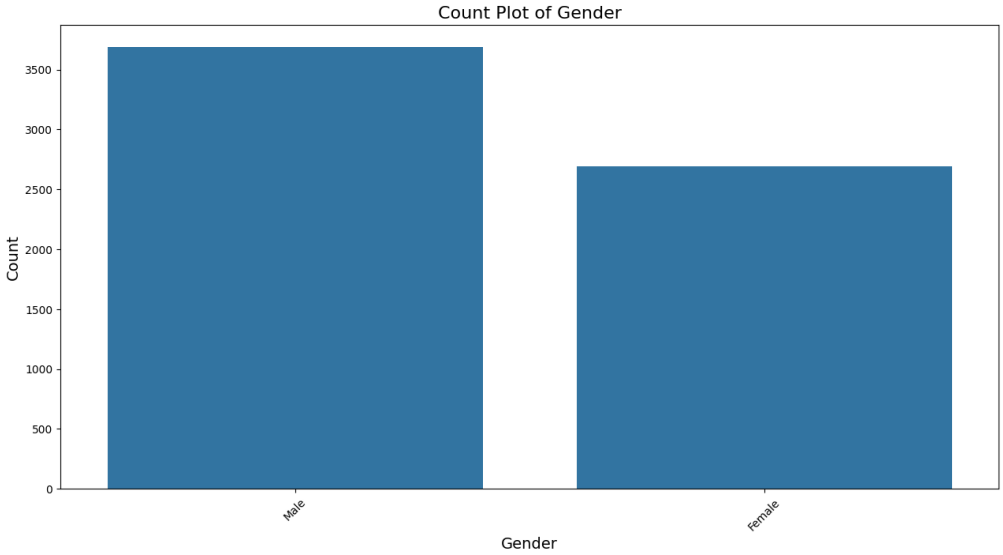


Figure 19: Gender count

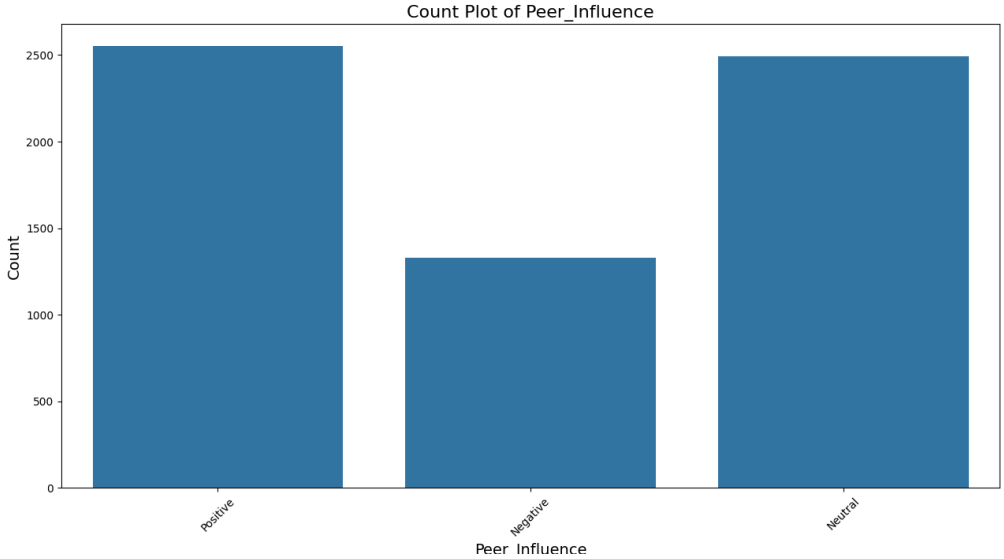


Figure 20: Peer influence count

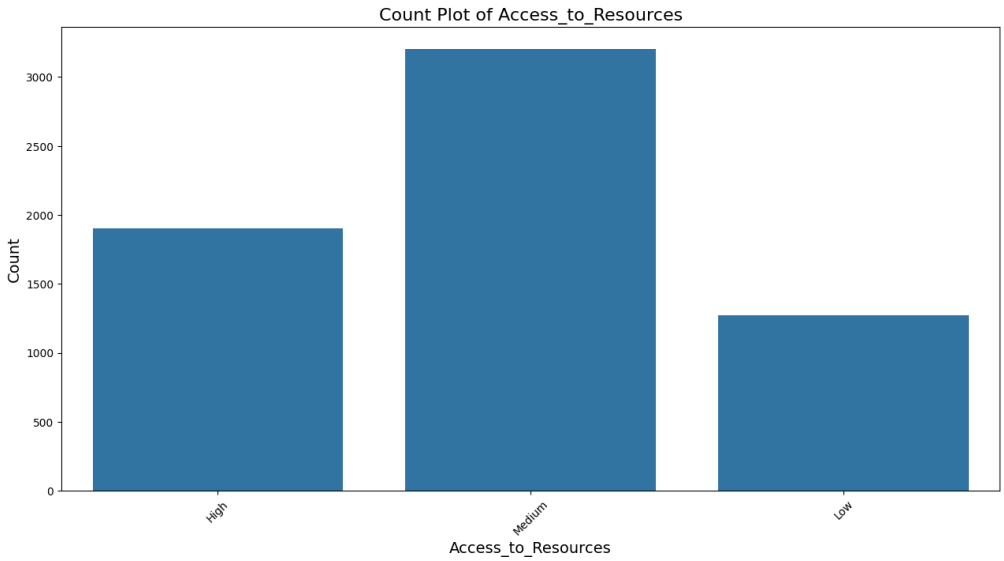


Figure 21: Resources count

Resources, peer influence, gender, family income and extracurricular activity are fairly balanced.